

CS121P Introduction to Computer Programming Test #2

This Test is open-book (course textbook) and open-notes (your own notes from this class) and open-laptop, but closed-neighbor. You may not communicate with anyone during the Test, other than the Instructor. There are a total of 100 points possible.

Relax, take a deep breath, and pray ☺.

Your grade will be based upon both the correctness and quality of your solutions. Don't get stuck on one portion of the test, but look all of it over first. If you cannot completely solve something, do your best, and submit whatever you are able to, so that I will have more opportunity to award any applicable partial credit.

DO NOT LEAVE THE ROOM until I verify to you that your submission has been received successfully.

For the IN CLASS PORTION of this Test, implement solutions to ANY ONE (your choice) of the following problems. If you choose to implement solutions to more than one of these problems, you will receive normal credit for the one you do the best, plus a little bit of extra credit for each additional one of sufficient quality.

For the TAKE HOME PORTION of this Test, implement solutions to ANY TWO of the following problems.

(continued on next page)

1. Modify Homework 6 to choose randomly between a comma-separated list of choices for each substitution, as shown below.

```
$ python3 Test2_1.py
Enter (^madlib^): The (^noun^) is (^adjective^).
Enter number of blank lines to hide the madlib: 1
```

```
Enter comma-separated choices for noun: moose, lion, hammerhead shark, rabbit
Enter comma-separated choices for adjective: scary, hairy, quite contrary
The hammerhead shark is hairy.
```

```
$ python3 Test2_1.py
Enter (^madlib^): The (^noun^) is (^adjective^).
Enter number of blank lines to hide the madlib: 2
```

```
Enter comma-separated choices for noun: moose, lion, hammerhead shark, rabbit
Enter comma-separated choices for adjective: scary, hairy, quite contrary
The moose is quite contrary.
```

2. Modify Homework 8 to instead make the opposite transformation, starting with the “classes.txt” file as input and producing the “students.txt” file as output, as shown below.

```
$ python3 Test2_2.py
Reading input file "classes.txt"...
Error: classes.txt does not exist or it can't be opened for input.
Program exiting now...
```

```
$ python3 Test2_2.py
Reading input file "classes.txt"...
Input processed.
Writing output file "students.txt"...
Error: students.txt can't be opened for output.
Program exiting now...
```

```
$ python3 Test2_2.py
Reading input file "classes.txt"...
Input processed.
Writing output file "students.txt"...
Output processed.
Program exiting now...
```

3. Modify Homework 9 as shown below.

```
$ python3 Test2_3.py
*****
Printing main() source code:
*****
def main():
    print("'" * 30 + "\nPrinting main() source code:\n" + "'" * 30)
    import inspect
    print(str(inspect.getsource(main)))
    print("'" * 30 + "\nPrinting main() source output:\n" + "'" * 30)
    f1 = Fraction(-10, 2)
    f2 = Fraction(-10, -3)
    f3 = Fraction(10, -4)
    f4 = Fraction(10, 5)
    f5 = f1 - f2 * f3 / f4

    print("f1: " + str(f1))
    print("f2: " + str(f2))
    print("f3: " + str(f3))
    print("f4: " + str(f4))
    print("f5: " + str(f5))
    print("isinstance(f5, Fraction): " + str(isinstance(f5, Fraction)))
    print("isinstance(f5, MixedFraction): " + str(isinstance(f5, MixedFraction)))

    m1 = MixedFraction(f1)
    m2 = MixedFraction(f2)
    m3 = MixedFraction(f3)
    m4 = MixedFraction(f4)
    m5 = m1 - m2 * m3 / m4

    print("m1: " + str(m1))
    print("m2: " + str(m2))
    print("m3: " + str(m3))
    print("m4: " + str(m4))
    print("m5: " + str(m5))
    print("isinstance(m5, Fraction): " + str(isinstance(m5, Fraction)))
    print("isinstance(m5, MixedFraction): " + str(isinstance(m5, MixedFraction)))

    print("(f5 == m5) : " + str(f5 == m5))
    print("(f5 != m5) : " + str(f5 != m5))
    print("(f1 < m2) : " + str(f1 < m2))
    print("(f2 <= m3) : " + str(f2 <= m3))
    print("(f3 > m4) : " + str(f3 > m4))
    print("(f4 >= m5) : " + str(f4 >= m5))

*****
Printing main() source output:
*****
f1: -5/1
f2: 10/3
f3: -5/2
f4: 2/1
f5: -5/6
isinstance(f5, Fraction): True
isinstance(f5, MixedFraction): False
m1: -5_(0/1)
m2: 3_(1/3)
m3: -2_(1/2)
m4: 2_(0/1)
m5: -0_(5/6)
isinstance(m5, Fraction): True
isinstance(m5, MixedFraction): True
(f5 == m5) : True
(f5 != m5) : False
(f1 < m2) : True
(f2 <= m3) : False
(f3 > m4) : False
(f4 >= m5) : True
```

~~~ *END OF CS121P TEST #2* ~~~